# Egg segmentation and tracking

Computer Vision Intern Task | reproducible experimental report

## Executive summary

A pretrained MobileSAM model produced reference egg masks for all 110 images. A classical local graph-cut method, selected on the development set, achieved mean IoU **0.565** and Dice **0.648** on 22 evaluation images. A two-ID tracker processed the complete 60-second reference video and marks uncertain reidentification after occlusion.

## Dataset and split

Roboflow egg v1 classification export: 110 images (108 in the egg folder and 2 Unlabeled), 640 x 640, CC BY 4.0. The supplied ZIP is v1; the online v2/v3 pages list 160 images but were not the provided local export. A deterministic 4:1 split gives 88 development and 22 evaluation images. pHash and geometrically verified ORB matches form 103 scene groups; all members of a group remain on one side of the split.



The provided images span hands, ash, cookware, fabric and patterned backgrounds.

## Task 1 - pretrained reference segmentation

MobileSAM automatic proposals at 384 px were filtered by area, shape, border contact and overlap. Visual audit found missed eggs, so 23 images received human-selected positive prompts; four difficult images used boxes. Prompts were chosen from the image, independent of classical predictions. Two boiling-water images have no visible egg mask. All masks, prompt coordinates and model checksum are saved. These are **pseudo-labels**, not human ground truth; agreement with them does not prove absolute accuracy.

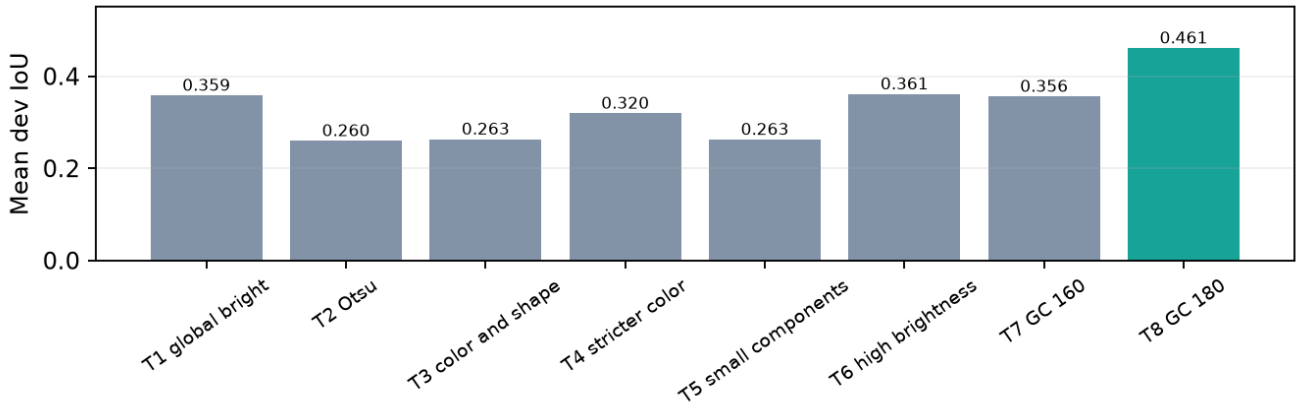
## Task 2 - classical method

The final method converts each frame to grayscale and HSV, selects bright low-saturation seed regions, removes tiny noise, filters implausible components by shape, and runs local GrabCut around each seed. GrabCut fits foreground/background color models and expands a seed toward the full egg boundary. The output is a union of candidate masks. The video uses a documented stricter brightness setting and table ROI; it does not use MobileSAM at runtime.

# Experiment log

Each trial used the same 88-image development set. Parameters and reasons are in outputs/experiment_log.json; the table below gives the decision path. The selection rule was highest mean dev IoU.

| Trial | Change / reason | Dev IoU | Decision |
|---|---|---|---|
| T1 global bright | Simple light-egg baseline | 0.359 | Drop |
| T2 Otsu | Adaptive global threshold for varying exposure | 0.260 | Drop |
| T3 color and shape | Reject colored background and non-elliptical regions | 0.263 | Drop |
| T4 stricter color | Reduce bright clutter false positives | 0.320 | Drop |
| T5 small components | Recover distant eggs | 0.263 | Drop |
| T6 high brightness | Test stronger foreground contrast | 0.361 | Drop |
| T7 graph cut 160 | Grow bright egg cores to full boundaries using local color models | 0.356 | Drop |
| T8 graph cut 180 | Stricter seeds before local graph cut | 0.461 | Keep |

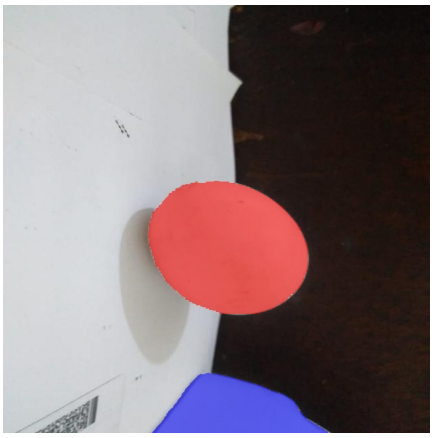



## Task 3 - held-out evaluation

Mean IoU 0.565 (image-bootstrap 95% interval 0.416-0.705); Dice 0.648; pixel precision 0.712; pixel recall 0.677. Object matching at IoU ≥ 0.5 yields precision 0.644 and recall 0.600; mean absolute count error 1.00. Metrics are per-image macro means; the bootstrap interval reflects small-sample variation, not label uncertainty.

# Qualitative error analysis

Green = overlap; red = MobileSAM only; blue = classical method only. Samples are the lowest-IoU evaluation cases, chosen by a fixed ranking. Open outputs/explorer/index.html for the full interactive gallery.



not_damaged_30_jpg
IoU 0.000



not_damaged_135_jpg
IoU 0.000



not_damaged_150_jpg
IoU 0.000

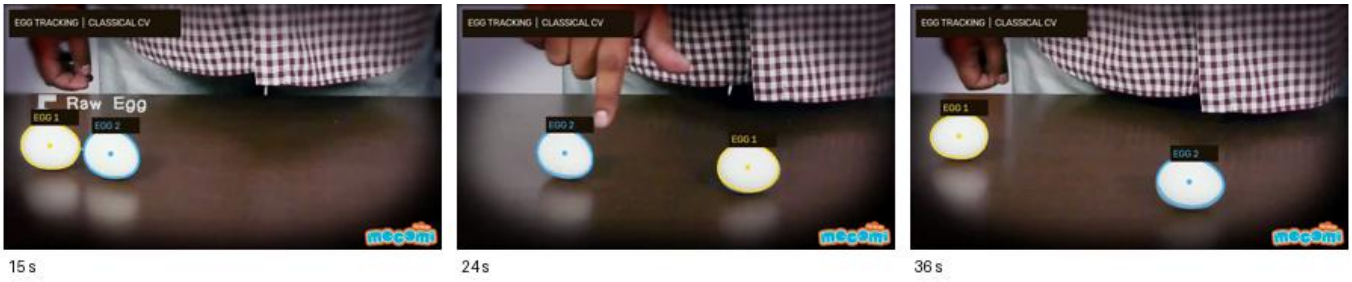
The first case selects bright tissue while missing the egg. The second is an Unlabeled boiling-water frame: a reflection is falsely predicted as an egg. The third misses a shadowed egg entirely. These cases expose different failure mechanisms rather than one generic threshold error.

## Observed gaps

Dark or shadowed eggs can fail the bright-seed step. Bright cookware, tissue and reflections can become false positives. GrabCut can absorb a similar-colored background or split an egg occluded by a rail. The reference pipeline itself misses partially submerged eggs and sometimes selects non-egg proposals; the saved QC sheet and manual prompt log make this visible. Next improvements: human instance masks for a representative subset, illumination normalization, temporal consistency for video, and a dedicated egg detector to provide proposals.

# Video tracking and reproducibility

Task 4 uses a table ROI and a stricter brightness seed to suppress hands, captions and reflections. Connected components are assigned to two persistent tracks with global minimum-distance matching; a frame gap over 24 frames or a jump over 100 px is marked uncertain. Across 1,440 video frames, the output contains 1351 egg detections and 3 uncertain reidentification events. Identity during hand occlusion cannot be established confidently from two visually similar eggs; these events are explicitly signalled rather than silently relabelled.




Three frames from the annotated output. IDs remain distinct; an uncertain recovery is marked with ?.

## Tracking video

Local review video: outputs/tracked_eggs.mp4. Public Drive publication is pending approval.

## Reproduce

Create a Python 3.12 environment, install requirements.txt, keep the supplied egg.v1i.folder at the repository root, and run scripts/generate_reference.py, scripts/repair_reference.py, scripts/run_experiments.py, scripts/track_video.py, and scripts/make_report.py in that order. See README.md for exact commands. The MobileSAM checkpoint is downloaded separately and identified by checksum in outputs/reference_provenance.json.

## Attribution and AI use

Dataset: Fire, egg v1, Roboflow Universe, CC BY 4.0, https://universe.roboflow.com/fire-bf7qe/egg-vzhtr/dataset/1 . Model: Ultralytics MobileSAM, https://docs.ultralytics.com/models/sam/ . Source video: Mocomi, https://www.youtube.com/watch?v=Y4JyV94y2_A . AI coding assistance was used for code drafting, debugging, and report wording; the workflow, visual QC, method comparison, and interpretation are documented for independent review.

## Limitations

The dataset contains only classification folders, not segmentation annotations. The same images were inspected during development, so the reported agreement is exploratory and should be confirmed with independent human-labelled masks. Scene grouping reduces but cannot eliminate leakage from visually related images. The video-specific ROI is tailored to this clip and should not be assumed to transfer to other scenes.